

Math 81: Abstract Algebra

Prishita Dharampal

Problem 1. *Construct it!*

- (a) Prove that if n and m are relatively prime positive integers such that a regular n -gon and a regular m -gon are constructible, then a regular nm -gon is constructible. (In case you know it, you cannot use Gauss's classification of constructible n -gons.)
- (b) Prove that for any $b, c \in \mathbb{Q}$, any root of $x^{256} + bx^{128} + c$ is constructible.

Solution.

1. **Claim:** If an n -gon is constructible if and only if the n -th root of unity ζ_n is constructible.

Proof: ($\implies$)

If an n -gon is constructible, then its internal angle $\frac{2\pi}{n}$ is constructible. This angle is the angle required to construct ζ_n , since the angle subtended at the origin by ζ_n and ζ_n^2 is $\frac{2\pi}{n}$.

($\impliedby$)

If an ζ_n is constructible, then the angle subtended at the origin by ζ_n and ζ_n^2 is constructible, i.e. $\frac{2\pi}{n}$ is constructible. And from HW3 we know that if $\frac{2\pi}{n}$ is constructible then a regular n -gon is constructible.

Hence proved.

To show that a regular nm -gon is constructible we need to show that ζ_{nm} is constructible. Note that since $\gcd(n, m) = 1$ by Bezout's Identity we have

$$an + bm = 1, \quad a, b \in \mathbb{Z}$$

Now, since a regular n -gon and a regular m -gon are constructible then ζ_n and ζ_m are constructible.

$$\begin{aligned}
\zeta_{nm}^1 &= \zeta_{nm}^{an+bm} \\
&= \zeta_{nm}^{an} \cdot \zeta_{nm}^{bm} \\
&= (\zeta_{nm}^n)^a \cdot (\zeta_{nm}^m)^b
\end{aligned}$$

But note that $\zeta_{nm}^n = \zeta_m$ and $\zeta_{nm}^m = \zeta_n$,

$$= (\zeta_m)^a \cdot (\zeta_n)^b$$

Since the constructible numbers are a field, $(\zeta_m)^a \cdot (\zeta_n)^b$ is constructible and hence so is ζ_{nm} . That is, we can construct a regular nm -gon.

2. Let $f(x) = x^{256} + bx^{128} + c$ over $\mathbb{Q}[x]$. And let K be the splitting field for this polynomial. Substituting $y = x^{128}$, we get $g(y) = y^2 + by + c$. Let α be a root to $g(y)$. Then we have the following two cases:

(a) $g(y)$ is reducible.

Then $g(y) = h(x)h'(x)$ in $\mathbb{Q}$, where $h(x), h'(x)$ are both linear polynomials, i.e. their roots are in $\mathbb{Q}$ and hence are constructible.

(b) $g(y)$ is irreducible.

Then $g(y)$ is the minimal polynomial for α and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2$. Since any degree two extensions of a constructible field are constructible, α is constructible. Moreover $g(y) = (y - \alpha)h(x)$ in $\mathbb{Q}(\alpha)$, where $h(x)$ is some linear polynomial, i.e. it's root is also in $\mathbb{Q}(\alpha)$ and hence is constructible.

That is over it's splitting field K_g $g(y) = (y - \alpha)(y - \beta)$, where α, β are it's roots. Substituting back, we get,

$$f(x) = (x^{128} - \alpha)(x^{128} - \beta)$$

Consider the polynomial, $f_\alpha(x) = x^{128} - \alpha$. Solving $x^{128} = \alpha$ is the same as repeatedly taking squareroots.

$$\begin{aligned}
x^{128} &= \alpha \\
x^{64} &= \pm \sqrt{\alpha} \\
x^{32} &= \pm \sqrt{\sqrt{\alpha}} \\
&\vdots \\
x^2 &= \pm \sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\alpha}}}}}}} \\
x &= \pm \sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\alpha}}}}}}}
\end{aligned}$$

Note, at each step we adjoin a square root and hence the degree of field extension (at each step) is at most **2**. Since we must take **7** square roots, $[K_\alpha : \mathbb{Q}(\alpha)] \leq 2^7$. And since at each step adjoining a root is at most a degree two extension over a constructible field, hence all roots of $f_\alpha(x)$ are constructible.

Similarly, the roots of the polynomial $f_\beta = x^{128} - \beta$ give us a tower of degree two extensions such that it's splitting field K_β is atmost a 2^7 degree extension over $\mathbb{Q}(\beta)$. And since at each step adjoining a root is at most a degree two extension over a constructible field, hence all roots of $f_\beta(x)$ are constructible.

Hence, all roots of $x^{256} + bx^{128} + c$ lie in fields obtained by repeatedly adjoining square roots to $\mathbb{Q}$, and hence are constructible.

Problem 2. *Simple splitting.*

Let p be a prime number.

- (a) Prove that the splitting field of $x^p - 3$ has degree $p(p-1)$ over $\mathbb{Q}$.
- (b) Find a simple generator for the splitting field of $x^3 - p$ over $\mathbb{Q}$.

Solution.

Claim: For p prime, all p roots of the polynomial $f(x) = x^p - \alpha$, are distinct and of the form $\zeta_p^k \cdot \sqrt[p]{\alpha}$, where ζ_p is the p -th root of unity, and $k \in \{1, \dots, p\}$.

Proof: Note that $\sqrt[p]{\alpha}$ is a root,

$$f(\sqrt[p]{\alpha}) = (\sqrt[p]{\alpha})^p - \alpha = \alpha - \alpha = 0$$

We prove the claim by showing that $\forall k \in \{1, \dots, p\}$, $\zeta_p^k \cdot \sqrt[p]{\alpha}$ gives us p distinct roots of $f(x)$.

$$\begin{aligned} f(\zeta_p^k \cdot \sqrt[p]{\alpha}) &= (\zeta_p^k \cdot \sqrt[p]{\alpha})^p - \alpha \\ &= (\zeta_p^k)^p (\sqrt[p]{\alpha})^p - \alpha \\ &= \zeta_p^{kp}(\alpha) - \alpha \\ &= (1)^k(\alpha) - \alpha \quad (\zeta_p^p = 1) \\ &= \alpha - \alpha = 0 \end{aligned}$$

That is, $\forall k \in \{1, \dots, n\}$, $\zeta_p^k \cdot \sqrt[p]{\alpha}$ is a root.

This gives us exactly p roots because for $i \neq j$, $\zeta_p^i \cdot \sqrt[p]{\alpha} = \zeta_p^j \cdot \sqrt[p]{\alpha} \implies \zeta_p^i = \zeta_p^j$, since $\sqrt[p]{\alpha} \neq 0$, and primitive p -th roots of unity are distinct.

1. By the claim, we know that the roots of this polynomial are of the form $\zeta_p^k \cdot \sqrt[p]{3}$, for $k \in \{1, \dots, p\}$. Thus, the splitting field $K = \mathbb{Q}(\sqrt[p]{3}, \zeta_p)$.

Note that $x^p - 3$ is irreducible over $\mathbb{Q}$ by Eisenstein and it is the minimal polynomial for $\sqrt[p]{3}$.

$$[\mathbb{Q}(\sqrt[p]{3}) : \mathbb{Q}] = p$$

Now, we have two cases:

- (a) $p = 2$ Then, $\zeta_p = -1$, and $-1 \in \mathbb{Q}[\sqrt{3}]$. That is

$$[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 2 = 2(2-1)$$

- (b) $p > 2$ Then, $\zeta_p \notin \mathbb{Q}(\sqrt[p]{3})$ since $\mathbb{Q}(\sqrt[p]{3}) \subset \mathbb{R}$, and ζ_p is not real for $p > 2$. From HW3, we know that

$$[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$$

Since $\mathbb{Q}(\zeta_p)$ is a subfield of K , by tower law,

$$[K : \mathbb{Q}] = [K : \mathbb{Q}(\zeta_p)] \cdot [\mathbb{Q}(\zeta_p) : \mathbb{Q}]$$

and,

$$[K : \mathbb{Q}(\zeta_p)] \leq p$$

That is, $[K : \mathbb{Q}] \leq p(p - 1) = n$.

Since both $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\sqrt[p]{3})$ are subfields of K , their degrees, p , $(p - 1)$ must divide n . Since these two numbers are relatively prime it follows that n is divisible by $p(p - 1)$ so we must have

$$[\mathbb{Q}(\sqrt[p]{3}, \zeta_p) : \mathbb{Q}] = p \cdot (p - 1)$$

Hence proved.

2. Consider $f(x) = x^3 - p \in \mathbb{Q}[x]$.

Again by claim, we know all the roots of $f(x)$ are of the form $\zeta_3^k \cdot \sqrt[p]{p}$, $k \in \{1, 2, 3\}$. That is the splitting field of $f(x)$ is $\mathbb{Q}(\zeta_3, \sqrt[p]{p})$.

A simple extension for this field is $\mathbb{Q}(\zeta_3 + \sqrt[p]{p})$. The containment $\mathbb{Q}(\zeta_3 + \sqrt[p]{p}) \subset \mathbb{Q}(\zeta_3, \sqrt[p]{p})$ is obvious. To show the reverse containment $\mathbb{Q}(\zeta_3, \sqrt[p]{p}) \subset \mathbb{Q}(\zeta_3 + \sqrt[p]{p})$, we check

$$\begin{aligned} (\zeta_3 + \sqrt[p]{p})^3 &= 1 + p + 3(\zeta_3^2 \sqrt[p]{p} + \zeta_3 (\sqrt[p]{p})^2) \\ &= 1 + p + 3\zeta_3 \sqrt[p]{p} (\zeta_3 + \sqrt[p]{p}) \\ &= 1 + p + 3\zeta_3 \sqrt[p]{p} (\zeta_3 + \sqrt[p]{p}) \\ \frac{(\zeta_3 + \sqrt[p]{p})^3 - 1 - p}{3(\zeta_3 + \sqrt[p]{p})} &= \zeta_3 \sqrt[p]{p} \end{aligned}$$

That is, $\zeta_3 \sqrt[p]{p} \in \mathbb{Q}(\zeta_3 + \sqrt[p]{p})$

Then we get, $\zeta_3 \sqrt[p]{p} (\zeta_3 + \sqrt[p]{p}) = \frac{(\zeta_3 + \sqrt[p]{p})^3 - 1 - p}{3} = \zeta_3^2 \sqrt[p]{p} + \zeta_3 (\sqrt[p]{p})^2$. Note that $\zeta_3 + \zeta_3^2 + 1 = 0$

$$\begin{aligned}
\frac{(\zeta_3 + \sqrt[3]{p})^3 - 1 - p}{3} &= \zeta_3^2 \sqrt[3]{p} + \zeta_3 (\sqrt[3]{p})^2 \\
\frac{(\zeta_3 + \sqrt[3]{p})^3 - 1 - p}{3} &= (-1 - \zeta_3) \sqrt[3]{p} + \zeta_3 (\sqrt[3]{p})^2 \\
\frac{(\zeta_3 + \sqrt[3]{p})^3 - 1 - p}{3} &= -\sqrt[3]{p} - \zeta_3 \sqrt[3]{p} + \zeta_3 (\sqrt[3]{p})^2 \\
\frac{(\zeta_3 + \sqrt[3]{p})^3 - 1 - p}{3} + \zeta_3 \sqrt[3]{p} &= -\sqrt[3]{p} + \zeta_3 (\sqrt[3]{p})^2 \\
\frac{(\zeta_3 + \sqrt[3]{p})^3 - 1 - p}{3} + \zeta_3 \sqrt[3]{p} &= \sqrt[3]{p} (\zeta_3 \sqrt[3]{p} - 1) \\
\left(\frac{(\zeta_3 + \sqrt[3]{p})^3 - 1 - p}{3} + \zeta_3 \sqrt[3]{p} \right) \cdot \left(\frac{1}{\zeta_3 \sqrt[3]{p} - 1} \right) &= \sqrt[3]{p}
\end{aligned}$$

If $\sqrt[3]{p}$ is in $\mathbb{Q}(\zeta_3 + \sqrt[3]{p})$ then so is ζ_3 . Hence, $\mathbb{Q}(\zeta_3 + \sqrt[3]{p}) = \mathbb{Q}(\zeta_3, \sqrt[3]{p})$.

Problem 3. Embeddings.

For each triple of fields F , K , and L , calculate the size of $\text{Hom}_F(K, L)$.

- (a) $F = \mathbb{Q}$, $K = \mathbb{Q}(\sqrt{2})$, $L = \mathbb{Q}(\alpha)$ where α is a root of $x^4 - 2x^2 - 1$.
- (b) $F = \mathbb{Q}$, $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $L = \mathbb{Q}(\sqrt{\sqrt{2} + \sqrt{3}})$.
- (c) $F = \mathbb{Q}$, $K = \mathbb{Q}[x]/(x^3 + x - 1)$, $L = \mathbb{R}$.
- (d) $F = \mathbb{F}_2$, $K = \mathbb{F}_4$, $L = \mathbb{F}_8$.

Solution.

Note that (by FT Proposition 2.4) we know that if K is a simple extension of F given by the minimal polynomial $f(x) \in F[x]$, L is another extension of F and $\varphi : K \rightarrow L$ is a homomorphism then there is a bijection between roots of $f(x)$ in L and homomorphisms from K to L .

1. The minimal polynomial of $\sqrt{2}$ is $f(x) = x^2 - 2$ over $F[x]$.

Let α is a root to $x^4 - 2x^2 - 1$, substituting $y = x^2$ we get,

$$\begin{aligned}
 y^2 - 2y - 1 &= 0 \\
 y^2 - 2y + 1 - 2 &= 0 \\
 \implies (y - 1)^2 - (\sqrt{2})^2 &= 0 \\
 \implies (x^2 - 1 - \sqrt{2})(x^2 - 1 + \sqrt{2}) &= 0
 \end{aligned}$$

The roots of this polynomial look like $\pm\sqrt{1 \pm \sqrt{2}}$. Since α is a root to this polynomial, α^2 is either $1 + \sqrt{2}$ or $1 - \sqrt{2}$. That is, $\sqrt{2}$ is in the field $\mathbb{Q}(\alpha)$, and hence, so is $-\sqrt{2}$.

Since both roots of $f(x)$ are in L , $|\text{Hom}_F(K, L)| = 2$

2. $f(x) = (x^2 - 2)(x^2 - 3)$

Note that $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$ since the containment $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is trivial and the reverse containment can be shown as follows:

$$\begin{aligned}
 \sqrt{2} + \sqrt{3} - (\sqrt{2} + \sqrt{3})^{-1} &= \sqrt{2} + \sqrt{3} - \frac{\sqrt{2} - \sqrt{3}}{2 - 3} \\
 &= \sqrt{2} + \sqrt{3} - \sqrt{2} + \sqrt{3} \\
 &= 2\sqrt{3}
 \end{aligned}$$

That is $\sqrt{3} \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$ and then $\sqrt{2} \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$.

Hence, $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$

Note that $(\sqrt{\sqrt{2} + \sqrt{3}})^2 = \sqrt{2} + \sqrt{3}$, so $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subset K$, and all four roots of $f(x)$ are in K , so $|\text{Hom}_F(K, L)| = 4$

3. $f(x) = x^3 + x - 1$

Note that $f(x)$ is irreducible over F by the rational root test. Let α be a root of $f(x)$, then (from HW4) the splitting field of $f(x)$ is $\mathbb{Q}(\alpha, \sqrt{\Delta f})$ where $\Delta(f) = -31$

Since $\mathbb{Q}(\alpha) \subset \mathbb{Q}(\alpha, \sqrt{\Delta f})$, the other two roots must have a complex component. And α cannot have a complex component, because complex roots come in pairs and $f(x)$ only has 3 roots. Hence, only $\alpha \in L$, and $|\text{Hom}_F(K, L)| = 1$.

4. $|\text{Hom}_F(K, L)| = 0$

From Problem 4, we know that $[\mathbb{F}_4 : \mathbb{F}_2] = 2$ and $[\mathbb{F}_8 : \mathbb{F}_2] = 3$, by the tower law there are no possible embeddings of $\mathbb{F}_4$ in $\mathbb{F}_8$.

Problem 4. *Finite fields.*

- (a) Let p be a prime and $q = p^n$. Prove that $\mathbb{F}_q/\mathbb{F}_p$ is a Galois extension, whose Galois group is cyclic of order n , generated by the Frobenius $\varphi : x \mapsto x^p$.
- (b) Calculate the number of elements $\alpha \in \mathbb{F}_{16}$ such that $\mathbb{F}_{16} = \mathbb{F}_2(\alpha)$.
- (c) Consider the splitting field $K/\mathbb{F}_7$ of the polynomial $f(x) = x^3 - 2 \in \mathbb{F}_7[x]$. Compute the order $|K|$, find some $\alpha \in K$ such that $K = \mathbb{F}_7(\alpha)$, and find all the roots of $f(x)$ in K as polynomials in α . (To ponder: what is the analogue of ω ?)

Solution.

- (a) We know a field extension K/F is Galois if K is the splitting field over F of a separable polynomial $f(x)$ (Corollary 6). And we also know that a polynomial $f(x)$ is separable if and only if $\gcd(f(x), f'(x)) = 1$, that is they share no roots (where $f'(x)$ is the derivative of $f(x)$).

Consider the polynomial $f(x) = x^q - x$ over the field $\mathbb{F}_p[x]$.

$$f'(x) = qx^{q-1} - 1 = -1$$

Since, $q = p^n$ and $p \equiv 0$ in $\mathbb{F}_p$, $f'(x)$ has no roots, hence $f(x)$ is separable.

The splitting field K of $f(x)$ has exactly q distinct roots of $f(x)$. Let $S \subset K$ be the set of roots of the polynomial $f(x)$.

$$|S| = q$$

Note that for any $\alpha, \beta \in S$, $\alpha^q = \alpha$, $(\alpha\beta)^q = \alpha\beta$, $(\alpha^{-1})^q = \alpha^{-1}$, and (by Freshman's Dream), $(\alpha + \beta)^q = \alpha + \beta$. That is S is a subfield of K containing all the roots and by minimality of splitting fields, $S = K$. That is $|K| = q = p^n$. Since there is a unique finite field of order q up to isomorphism, we write $K = \mathbb{F}_q$. Then,

$$[\mathbb{F}_q : \mathbb{F}_p] = n$$

Then, (by Corollary 6) $\mathbb{F}_q/\mathbb{F}_p$ is a Galois extension.

The Galois group $G = \text{Aut}_{\mathbb{F}_p}(\mathbb{F}_q)$ of this extension has order n . Consider the Frobenius map $\phi : \mathbb{F}_q \rightarrow \mathbb{F}_q$.

$$\phi(x) = x^p$$

Note that for any $\alpha, \beta \in \mathbb{F}_q$, the following hold:

- (a) $\phi(\alpha + \beta) = \alpha^p + \beta^p$ (by Freshman's Dream) $= \phi(\alpha) + \phi(\beta)$.
- (b) $\phi(\alpha\beta) = \alpha^p\beta^p = \phi(\alpha)\phi(\beta)$

Hence, ϕ is a homomorphism. Moreover, we know that any field homomorphism is injective, and since the map is defined from $\mathbb{F}_q \rightarrow \mathbb{F}_q$, it is also surjective and hence an automorphism. Also, $\forall x \in \mathbb{F}_p \subset \mathbb{F}_q$, $\phi(x) = x^p = x$, i.e. ϕ fixes the field $\mathbb{F}_p$.

$$\phi \in \text{Aut}_{\mathbb{F}_p}(\mathbb{F}_q)$$

Now consider $\phi^2 \in \text{Aut}_{\mathbb{F}_p}(\mathbb{F}_q)$, for any $x \in \mathbb{F}_q$, $\phi^2(x) = \phi(\phi(x)) = x^{p^2}$. Similarly, for all $i = 1, 2, \dots, n$

$$\phi^i(x) = x^{p^i}$$

and only $\phi^n(x) = x^{p^n} = x$.

Suppose the order of ϕ is not n , then $|\phi| = i$, where $i < n$. Then

$$x^{p^i} = x, \forall x \in \mathbb{F}_q$$

that is, all elements of $\mathbb{F}_q$ are solutions to $x^{p^i} - x$. But that polynomial has degree p^i and would have at most p^i roots but since $p^i < q$ this is a contradiction. So, $|\phi| = n$. That is $\langle \phi \rangle$ generates a subgroup of $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_q)$ of order n , but since $\mathbb{F}_q/\mathbb{F}_p$ is Galois, $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_q) = n$, hence $\langle \phi \rangle$ is a subgroup of $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_q)$ of equal order and must be the whole group.

Hence, $\mathbb{F}_q/\mathbb{F}_p$ is a Galois extension, whose Galois group is cyclic of order n , generated by the Frobenius $\phi : x \mapsto x^p$.

(b) $\mathbb{F}_{16} = \mathbb{F}_2(\alpha)$

Note that $16 = 2^4$, then by part (a) we know that

$$|\text{Aut}_{\mathbb{F}_2}(\mathbb{F}_{16})| = [\mathbb{F}_{16} : \mathbb{F}_2] = 4$$

So, $\mathbb{F}_{16}$ must be generated by the roots of a quartic polynomial over $\mathbb{F}_2$. Again from part (a) we know that all 16 elements in $\mathbb{F}_{16}$ are roots to this polynomial. However, note that it is not necessarily true that the polynomial is minimal for all 16 elements.

By proposition 15, we know that the subfields of $\mathbb{F}_{16}$ are Galois over $\mathbb{F}_2$ and are in one-to-one correspondence with all divisors d of 4, i.e., $d = 1, 2, 4$. That is, they are the fields $\mathbb{F}_{2^d}$ which are the fixed fields of ϕ^d .

The possible values for $\mathbb{F}_{2^d}$ are $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_{16}$ with 2, 4, 16 elements respectively. And,

$$\mathbb{F}_2 \subset \mathbb{F}_4 \subset \mathbb{F}_{16}$$

That is, there are $12(16 - 4)$ elements for which the quartic polynomial is minimal, and hence are generators for the field $\mathbb{F}_{16}$.

(c) Let K be the splitting field of the polynomial $x^3 - 2$ over $\mathbb{F}_7[x]$. Note that 2 is not a cube in $\mathbb{F}_7$, and hence $f(x)$ is irreducible (degree 3). Let α be a root so $\alpha^3 = 2 \pmod{7} \in K$, then $[\mathbb{F}_7(\alpha) : \mathbb{F}_7] = 3$. From part (a), $\mathbb{F}_7(\alpha) = \mathbb{F}_{7^3}$, so $|\mathbb{F}_7(\alpha)| = 7^3 = 343$.

Also, note that $K = \mathbb{F}_7(\alpha)$ is the splitting field because 1 is a cube in $\mathbb{F}_7$, for 1, 2, and 4. These are the analogues of ω in K . That is the roots of $f(x)$ in K are $\alpha, 2\alpha, 4\alpha$.

Problem 5. Dihedral.

Let F be a field of characteristic $\neq 2$ and let $a, b \in F$. Let $f(x) = x^4 + ax^2 + b \in F[x]$. Assume that $f(x)$ is separable. Let K be the splitting field of $f(x)$ over F . Prove that the Galois group of K/F is isomorphic to a subgroup of the dihedral group D_8 of order 8.

Solution.

Let $y = x^2$, then,

$$g(y) = y^2 + ay + b$$

Let the roots of this polynomial be p, q . Then $x^2 = p$, and $x^2 = q$, so the roots of $f(x)$ are $\pm\sqrt{p}, \pm\sqrt{q}$.

Also, note that for any root α of $f(x)$, $-\alpha$ is also a root. So, the polynomial would never factor into an irreducible cubic. Then, we have the following cases:

1. $f(x)$ completely factors over F .

Then $[K : F] = 1$.

2. $f(x)$ reduces to one irreducible quadratic over F .

Then $[K : F] = 2$.

3. $f(x)$ reduces to two irreducible quadratics over F .

Then $[K : F] = 4$.

4. $f(x)$ is irreducible over F , and factors completely after we adjoin a root.

Then $[K : F] = 4$.

5. $f(x)$ is irreducible over F , and factors into two distinct irreducible quadratics after we adjoin a root.

Then $[K : F] = 8$.

Hence, the degree of the extension K/F is at most 8. Moreover, since $f(x)$ is separable, $|\text{Gal}(K/F)| = [K : F]$.

Also note that, all groups of orders 1, 2, 4 are isomorphic to a subgroup of D_8 .

$$\{e\} \cong \{e\}, \quad \mathbb{Z}_2 \cong \langle s \rangle, \quad \mathbb{Z}_4 \cong \langle r \rangle, \quad K_4 \cong \langle s, r^2 \rangle$$

That only leaves us to prove that case 5 has a Galois group that is isomorphic to D_8 . Let the roots of $f(x)$ in K be $\{\sqrt{p}, -\sqrt{p}, \sqrt{q}, -\sqrt{q}\}$. Let $\sqrt{p} = \alpha, \sqrt{q} = \beta$. Let $\sigma \in \text{Gal}(K/F)$. Since σ is a field automorphism, it must permute the roots and preserve algebraic relations. That is,

$$\sigma(-\alpha) = -\sigma(\alpha), \quad \sigma(-\beta) = -\sigma(\beta).$$

and, since $\alpha^2 = p$ and $\beta^2 = q$, applying σ gives

$$\sigma(\alpha)^2 = \sigma(p), \quad \sigma(\beta)^2 = \sigma(q).$$

but because p and q are roots to the same minimal polynomial over F , we must have

$$\sigma(p) \in \{p, q\}, \quad \sigma(q) \in \{p, q\}.$$

therefore,

$$\sigma(\alpha) \in \{\pm\alpha, \pm\beta\}, \quad \sigma(\beta) \in \{\pm\alpha, \pm\beta\},$$

such that the squaring them agrees with $\sigma(p)$ and $\sigma(q)$.

Then we can define the following automorphisms,

$$\begin{array}{ll} \sigma_{\text{id}} : \begin{cases} \alpha \mapsto \alpha \\ \beta \mapsto \beta \end{cases} & \sigma_4 : \begin{cases} \alpha \mapsto \beta \\ \beta \mapsto \alpha \end{cases} \\ \sigma_1 : \begin{cases} \alpha \mapsto -\alpha \\ \beta \mapsto \beta \end{cases} & \sigma_5 : \begin{cases} \alpha \mapsto -\beta \\ \beta \mapsto -\alpha \end{cases} \\ \sigma_2 : \begin{cases} \alpha \mapsto \alpha \\ \beta \mapsto -\beta \end{cases} & \sigma_6 : \begin{cases} \alpha \mapsto \beta \\ \beta \mapsto -\alpha \end{cases} \\ \sigma_3 : \begin{cases} \alpha \mapsto -\alpha \\ \beta \mapsto -\beta \end{cases} & \sigma_7 : \begin{cases} \alpha \mapsto -\beta \\ \beta \mapsto \alpha \end{cases} \end{array}$$

Note that this is all of them because they are 8 distinct automorphisms and $|\text{Gal}(K/F)| = 8$. Computing orders we get,

$$|\sigma_{\text{id}}| = 1, \quad |\sigma_1| = 2, \quad |\sigma_6| = 4$$

Also note that,

$$\begin{aligned} \sigma_2(x) &= \sigma_6^2(\sigma_1(x)) \\ \sigma_3(x) &= \sigma_6^2(x) \\ \sigma_4(x) &= \sigma_1(\sigma_6(x)) \\ \sigma_5(x) &= \sigma_6(\sigma_1(x)) \\ \sigma_7(x) &= \sigma_6^3(x) \end{aligned}$$

So, σ_1 and σ_6 are generators of the group. Let $\sigma_1 = s$, and $\sigma_6 = r$, then for $\text{Gal}(K/F)$ to be isomorphic to D_8 the relations $s^2 = r^4 = e$ and $rs = sr^{-1}$ must hold. Since we know the orders of σ_1 and σ_6 the first relation holds.

$$\begin{aligned}\sigma_6(\sigma_1(\alpha)) &= \sigma_6(-\alpha) = -\sigma_6(\alpha) = -\beta \\ \sigma_1(\sigma_6^{-1}(\alpha)) &= \sigma_1(-\beta) = -\beta\end{aligned}$$

and,

$$\begin{aligned}\sigma_6(\sigma_1(\beta)) &= \sigma_6(\beta) = -\alpha \\ \sigma_1(\sigma_6^{-1}(\beta)) &= \sigma_1(\alpha) = -\alpha\end{aligned}$$

The relations hold, and hence the groups are isomorphic.

Problem 6. *Galois dreams of eight.*

Let $K/\mathbb{Q}$ be the splitting field of the polynomial $x^4 + 1 \in \mathbb{Q}[x]$. Compute the degree $[K : \mathbb{Q}]$ and the Galois group $G = \text{Gal}(K/\mathbb{Q})$. Draw the lattice of subextensions of $K/\mathbb{Q}$, providing a simple generator for each, and the corresponding lattice of subgroups of G under the Galois correspondence.

Solution.

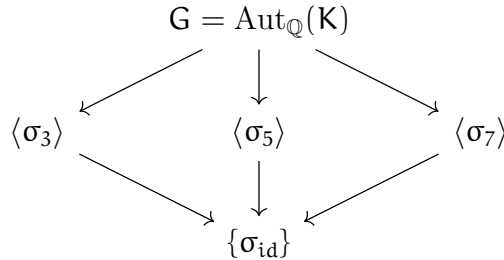
Consider the polynomial $x^8 - 1 = (x^4 - 1)(x^4 + 1)$. We know that the roots to this polynomial are the 8-th roots of unity, ζ_8 . Note that $\zeta_8^k = e^{2\pi i k/8}$. The even powers correspond to roots of $x^4 - 1$ since,

$$\zeta_8^k = e^{2\pi i j/4}$$

where $j = k/2$. Then the odd powers must be roots to $f(x) = x^4 + 1$. Since the roots are distinct, the polynomial is separable and the degree $[K : \mathbb{Q}] = 4$. Then, the group $G = \text{Aut}_{\mathbb{Q}}(K)$ also has order 4. The automorphisms are as follows,

$$\begin{aligned}\sigma_{\text{id}} : \zeta_8^i &\mapsto \zeta_8^i \\ \sigma_3 : \zeta_8^i &\mapsto \zeta_8^{3i} \\ \sigma_5 : \zeta_8^i &\mapsto \zeta_8^{5i} \\ \sigma_7 : \zeta_8^i &\mapsto \zeta_8^{7i}\end{aligned}$$

Subgroup Lattice:



Subextensions of $K/\mathbb{Q}$:

1. σ_3 fixes the subfield $\mathbb{Q}(\zeta_8^3 + \zeta_8)$.

$$\sigma_3(\zeta_8^3 + \zeta_8) = \sigma_3(\zeta_8^3) + \sigma_3(\zeta_8) = \zeta_8 + \zeta_8^3$$

$$\begin{aligned}
\zeta_8^3 + \zeta_8 &= \left(\cos\left(\frac{6\pi}{8}\right) + i \sin\left(\frac{6\pi}{8}\right) \right) + \left(\cos\left(\frac{2\pi}{8}\right) + i \sin\left(\frac{2\pi}{8}\right) \right) \\
&= \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right) + \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right) \\
&= -\left(\frac{\sqrt{2}}{2}\right) + i\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right) + i\left(\frac{\sqrt{2}}{2}\right) \\
&= 2i\left(\frac{\sqrt{2}}{2}\right) = i\sqrt{2}
\end{aligned}$$

2. σ_5 fixes the subfield $\mathbb{Q}(\zeta_8^2)$.

$$\sigma_5(\zeta_8^2) = \zeta_8^{10} = \zeta_8^2$$

3. σ_7 fixes the subfield $\mathbb{Q}(\zeta_8 + \zeta_8^7)$.

$$\sigma_7(\zeta_8 + \zeta_8^7) = \sigma_7(\zeta_8) + \sigma_7(\zeta_8^7) = \zeta_8^7 + \zeta_8$$

$$\begin{aligned}
\zeta_8 + \zeta_8^7 &= \left(\cos\left(\frac{2\pi}{8}\right) + i \sin\left(\frac{2\pi}{8}\right) \right) + \left(\cos\left(\frac{7\pi}{8}\right) + i \sin\left(\frac{7\pi}{8}\right) \right) \\
&= \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right) + \left(\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) \right) \\
&= \left(\left(\frac{\sqrt{2}}{2}\right) + i\left(\frac{\sqrt{2}}{2}\right) \right) + \left(\left(\frac{\sqrt{2}}{2}\right) - i\left(\frac{\sqrt{2}}{2}\right) \right) \\
&= \sqrt{2}
\end{aligned}$$

Subextension Lattice:

